

AGILE SPRINT PLANNING & USER STORY WORKSHOP

AI-Powered Smart Vocational Skill Assessment and Career Recommendation Platform

Assignment: CO1_AT2 : Agile Sprint Planning & User Story Workshop

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This document presents the complete Agile Sprint Planning and User Story Workshop for the AI-Powered Smart Vocational Skill Assessment and Career Recommendation Platform, organised into the following sections.

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Document Scope: Vision & stakeholders, 10 Epics with 27 user stories, Given–When–Then acceptance criteria, prioritised product backlog, Sprint 1 backlog with task breakdown, Fibonacci story-point estimation, and the Sprint 1 goal, deliverables and risk plan.

1. Project Vision and Stakeholders

1.1 Background

Educational institutions, training organisations and industries increasingly require a data-driven way to match learners with careers that suit their actual competencies rather than generic guidance. Traditional career counselling is often manual, one-size-fits-all and disconnected from real-time labour-market needs. The AI-Powered Smart Vocational Skill Assessment and Career Recommendation Platform is proposed to close this gap by combining online assessments, AI-driven aptitude analysis and live industry data into a single, personalised guidance system.

1.2 Project Vision

To build an intelligent, data-driven platform that empowers every student and job seeker to discover, prepare for, and pursue a career path aligned with their true skills, aptitude and interests — by continuously assessing competencies, benchmarking them against real industry demand, and guiding learners toward the right training, certification and job opportunities.

1.3 Project Objectives

- Assess technical and soft skills through structured, adaptive online assessments.
- Analyse aptitude and learning preferences using AI/ML models.
- Recommend personalised, ranked career pathways based on assessment outcomes.
- Match learner competencies against current industry and job-market requirements.
- Suggest relevant training and certification programs to close identified skill gaps.
- Generate personalised, shareable career reports for learners and counsellors.
- Monitor learning progress over time through a visual dashboard.
- Improve overall employability and workforce readiness through continuous, data-driven feedback.

1.4 Key Stakeholders

The table below identifies all stakeholders relevant to the platform, their role/interest in the project, and their relative influence on product decisions.

Stakeholder	Role / Interest in the Project	Influence
Students / Job Seekers (End Users)	Primary users who take assessments and receive career recommendations	High
Educational Institutions & Training Centres	Provide curriculum data, refer learners, integrate platform into counselling	High
Industry Partners / Employers	Supply live job-market and skill-demand data; post job/internship opportunities	High
Career Counsellors / Academic Advisors	Use AI insights to guide students; validate AI-generated recommendations	Medium
Certification & Training Providers	Offer courses/certifications suggested by the recommendation engine	Medium

Stakeholder	Role / Interest in the Project	Influence
Product Owner	Owns the product backlog, defines priorities and acceptance criteria	High
Scrum Master	Facilitates Agile ceremonies, removes impediments, tracks sprint progress	High
Development Team (AI/ML, Backend, Frontend, QA)	Designs, builds, tests and deploys the platform	High
Platform Administrators	Manage content, monitor system health, moderate data quality	Medium
Parents / Guardians (for minor students)	Support and monitor career decisions of school-level students	Low
Government / Skill-Development Bodies (e.g., NSDC)	Provide labour-market frameworks and standards for skill mapping	Medium
Data Privacy / Compliance Officer	Ensures assessment data and personal information meet privacy regulations	Medium

1.5 Success Criteria

- At least 80% of learners who complete assessments receive at least one career recommendation with a match score above 60%.
- Career recommendations are refreshed automatically whenever industry skill-demand data is updated.
- Learners can generate a complete, shareable career report in under 30 seconds.
- Counsellors report improved ability to guide students using AI-generated insights (measured via post-pilot survey).

2. Epics and User Stories

The product requirements were broken down into ten Epics, each representing a major functional capability of the platform. Each Epic was then decomposed into detailed User Stories following the standard Agile format: "As a <user>, I want <feature>, so that <benefit>." A total of 27 user stories were formulated across the ten epics, covering the full learner journey from onboarding to career report generation and progress tracking.

2.1 Epic Summary

Epic ID	Epic Title	Epic Goal
EP-01	User Onboarding & Profile Management	Enable learners to register, build a profile and manage their account securely.
EP-02	Technical & Soft Skill Assessment	Assess a learner's technical competencies and soft skills through adaptive online tests.
EP-03	AI-Based Aptitude & Learning Style Analysis	Use AI/ML models to analyse aptitude, cognitive strengths and preferred learning styles.

Epic ID	Epic Title	Epic Goal
EP-04	Career Path Recommendation Engine	Recommend personalised career pathways based on assessed skills, aptitude and interests.
EP-05	Industry Skill-Demand Matching	Compare learner competencies with live industry/job-market requirements.
EP-06	Training & Certification Recommendation	Suggest relevant courses and certification programs to close identified skill gaps.
EP-07	Personalized Career Report Generation	Generate a downloadable, shareable report summarising results and recommendations.
EP-08	Learning Progress Monitoring Dashboard	Track and visualise a learner's progress against recommended learning paths over time.
EP-09	Admin & Content Management	Allow administrators to manage assessment content, industry data feeds and platform quality.
EP-10	Notifications & Engagement	Keep learners engaged through reminders, alerts and progress notifications.

2.2 Detailed User Stories by Epic

EP-01: User Onboarding & Profile Management

US-01 As a new learner, I want to register using email or social login, so that I can quickly create an account and start my assessment.

Priority: Must | Business Value: 9/10 | Story Points: 3

US-02 As a learner, I want to build a profile with my education, interests and career goals, so that recommendations are personalised to me.

Priority: Must | Business Value: 8/10 | Story Points: 5

US-03 As a learner, I want to reset my forgotten password securely, so that I can regain access to my account.

Priority: Should | Business Value: 5/10 | Story Points: 2

EP-02: Technical & Soft Skill Assessment

US-04 As a learner, I want to take a technical skill assessment relevant to my domain, so that my current competency level is measured accurately.

Priority: Must | Business Value: 10/10 | Story Points: 8

US-05 As a learner, I want to take a soft-skill assessment (communication, teamwork, problem solving), so that non-technical strengths are also captured.

Priority: Must | Business Value: 9/10 | Story Points: 8

US-06 As a learner, I want the assessment to adapt its difficulty based on my answers, so that the test accurately reflects my true skill level.

Priority: Should | Business Value: 7/10 | Story Points: 13

US-07 As a learner, I want to save and resume an in-progress assessment, so that I do not lose my progress if I get interrupted.

Priority: Could | Business Value: 4/10 | Story Points: 5

EP-03: AI-Based Aptitude & Learning Style Analysis

US-08 As a learner, I want the system to analyse my aptitude using AI, so that I understand my natural strengths and reasoning ability.

Priority: Must | Business Value: 9/10 | Story Points: 13

US-09 As a learner, I want my preferred learning style (visual, auditory, kinesthetic) identified, so that suggested courses match how I learn best.

Priority: Should | Business Value: 6/10 | Story Points: 8

EP-04: Career Path Recommendation Engine

US-10 As a learner, I want to receive a ranked list of recommended career paths, so that I can explore options that fit my skills and interests.

Priority: Must | Business Value: 10/10 | Story Points: 13

US-11 As a learner, I want to see why a particular career was recommended to me, so that I can trust and understand the AI's reasoning.

Priority: Should | Business Value: 7/10 | Story Points: 8

US-12 As a learner, I want to compare two or more recommended career paths side by side, so that I can make an informed decision.

Priority: Could | Business Value: 5/10 | Story Points: 5

EP-05: Industry Skill-Demand Matching

US-13 As a learner, I want my skills compared against current industry/job requirements, so that I know how job-ready I am.

Priority: Must | Business Value: 9/10 | Story Points: 8

US-14 As an industry partner, I want to publish up-to-date skill requirements for open roles, so that learner recommendations reflect real market demand.

Priority: Must | Business Value: 8/10 | Story Points: 8

US-15 As a learner, I want to see the skill gap between my current profile and a target job role, so that I know exactly what to improve.

Priority: Should | Business Value: 7/10 | Story Points: 5

EP-06: Training & Certification Recommendation

US-16 As a learner, I want to receive suggested courses and certifications for my skill gaps, so that I can plan my learning path.

Priority: Must | Business Value: 8/10 | Story Points: 8

US-17 As a learner, I want to filter suggested courses by cost, duration and mode (online/offline), so that I can pick ones that fit my constraints.

Priority: Should | Business Value: 6/10 | Story Points: 5

US-18 As a training provider, I want my certification programs listed against relevant skill gaps, so that qualified learners can discover my courses.

Priority: Could | Business Value: 5/10 | Story Points: 5

EP-07: Personalized Career Report Generation

US-19 As a learner, I want to generate a personalised career report as a PDF, so that I can share it with counsellors, parents or employers.

Priority: Must | Business Value: 8/10 | Story Points: 5

US-20 As a career counsellor, I want to view a learner's full report with assessment and recommendation details, so that I can provide informed guidance.

Priority: Should | Business Value: 7/10 | Story Points: 5

EP-08: Learning Progress Monitoring Dashboard

US-21 As a learner, I want to track my learning progress on a dashboard, so that I stay motivated and aware of my improvement over time.

Priority: Should | Business Value: 7/10 | Story Points: 8

US-22 As a learner, I want to retake assessments periodically, so that my recommendations stay updated as my skills grow.

Priority: Could | Business Value: 5/10 | Story Points: 5

EP-09: Admin & Content Management

US-23 As an administrator, I want to add, edit or remove assessment questions, so that content stays accurate and current.

Priority: Must | Business Value: 7/10 | Story Points: 8

US-24 As an administrator, I want to monitor platform usage and assessment analytics, so that I can evaluate engagement and effectiveness.

Priority: Should | Business Value: 6/10 | Story Points: 5

US-25 As an administrator, I want to manage industry-partner data feeds, so that job-market information remains reliable.

Priority: Should | Business Value: 6/10 | Story Points: 5

EP-10: Notifications & Engagement

US-26 As a learner, I want to receive email/app notifications about incomplete assessments or new recommendations, so that I stay engaged with the platform.

Priority: Could | Business Value: 4/10 | Story Points: 3

US-27 As a learner, I want to receive alerts when a new job or certification matches my profile, so that I don't miss relevant opportunities.

Priority: Could | Business Value: 4/10 | Story Points: 5

3. Acceptance Criteria (Sprint 1 User Stories)

Acceptance criteria were written for every user story selected into Sprint 1, using the Given–When–Then format. This ensures each story has a clear, testable definition of what "done" means before development begins.

US-01: As a new learner, I want to register using email or social login, so that I can quickly create an account and start my assessment.

- Scenario 1: Given a new visitor on the registration page, When they enter a valid email and password (or choose social login) and submit, Then an account is created and a verification email is sent within 2 minutes.
- Scenario 2: Given an unverified account, When the learner clicks the verification link, Then the account status changes to 'Verified' and the learner is redirected to profile setup.

- Scenario 3: Given invalid input (e.g., malformed email, weak password), When the learner submits the form, Then inline validation errors are shown and no account is created.

US-02: As a learner, I want to build a profile with my education, interests and career goals, so that recommendations are personalised to me.

- Scenario 1: Given a verified learner, When they fill in education, interests and career-goal fields and save, Then the profile is stored and marked 'Complete'.
- Scenario 2: Given an incomplete profile, When the learner attempts to start an assessment, Then the system prompts them to complete mandatory profile fields first.
- Scenario 3: Given a saved profile, When the learner edits any field, Then the changes are updated and reflected immediately on the profile page.

US-04: As a learner, I want to take a technical skill assessment relevant to my domain, so that my current competency level is measured accurately.

- Scenario 1: Given a learner with a completed profile, When they start a technical skill assessment for their selected domain, Then a domain-relevant question set is presented with a visible timer.
- Scenario 2: Given an in-progress technical assessment, When the learner submits their final answer, Then a technical competency score is generated and saved to the learner's record.
- Scenario 3: Given a completed technical assessment, When the learner views results, Then scores are broken down by sub-skill (e.g., programming, data analysis).

US-05: As a learner, I want to take a soft-skill assessment (communication, teamwork, problem solving), so that non-technical strengths are also captured.

- Scenario 1: Given a learner with a completed profile, When they start the soft-skill assessment, Then scenario-based questions covering communication, teamwork and problem-solving are presented.
- Scenario 2: Given a completed soft-skill assessment, When results are generated, Then a soft-skill score with category-wise breakdown is displayed and stored.
- Scenario 3: Given a learner exits mid-assessment, When they return within 24 hours, Then they can resume from the last answered question.

US-08: As a learner, I want the system to analyse my aptitude using AI, so that I understand my natural strengths and reasoning ability.

- Scenario 1: Given completed technical and soft-skill assessment data, When the AI aptitude engine processes the results, Then an aptitude profile (logical, numerical, verbal reasoning) is generated within 60 seconds.
- Scenario 2: Given a generated aptitude profile, When the learner views their dashboard, Then strengths and improvement areas are displayed in plain language with a visual chart.
- Scenario 3: Given insufficient assessment data, When aptitude analysis is triggered, Then the system displays a message asking the learner to complete pending assessments.

US-10: As a learner, I want to receive a ranked list of recommended career paths, so that I can explore options that fit my skills and interests.

- Scenario 1: Given a learner with completed skill and aptitude analysis, When they open the 'Career Recommendations' page, Then a ranked list of at least five career paths is displayed with a match percentage.
- Scenario 2: Given a displayed career recommendation, When the learner clicks on it, Then a detail view shows required skills, growth outlook and average salary range.
- Scenario 3: Given no strong matches are found, When recommendations are generated, Then the system shows the closest matches along with guidance on skills to develop.

US-13: As a learner, I want my skills compared against current industry/job requirements, so that I know how job-ready I am.

- Scenario 1: Given a learner profile and live industry skill-demand data, When the matching engine runs, Then a job-readiness percentage is calculated for each recommended career path.
- Scenario 2: Given a job-readiness score below 60%, When the learner views the result, Then the top three missing skills are highlighted.
- Scenario 3: Given updated industry data, When it is refreshed in the system, Then existing learner match scores are recalculated within 24 hours.

US-16: As a learner, I want to receive suggested courses and certifications for my skill gaps, so that I can plan my learning path.

- Scenario 1: Given an identified skill gap, When the learner views the 'Recommended Courses' section, Then at least three relevant courses or certifications are listed, mapped to the specific gap.
- Scenario 2: Given a listed course, When the learner clicks 'View Details', Then provider, duration, cost and mode of delivery are displayed.
- Scenario 3: Given no matching course exists for a gap, When recommendations are generated, Then the system displays 'No course currently available' rather than an empty/broken page.

US-19: As a learner, I want to generate a personalised career report as a PDF, so that I can share it with counsellors, parents or employers.

- Scenario 1: Given completed assessments and recommendations, When the learner clicks 'Generate Report', Then a PDF report containing scores, aptitude summary, career matches and suggested courses is produced within 30 seconds.
- Scenario 2: Given a generated report, When the learner clicks 'Download' or 'Share', Then the report is downloaded locally or a shareable link is created.
- Scenario 3: Given a report generation failure, When an error occurs, Then the learner sees a clear error message with a retry option.

4. Product Backlog Prioritization

The product backlog was prioritised using a combination of MoSCoW categorisation (Must / Should / Could / Won't-this-sprint) and a Business Value score (1–10) assigned collaboratively by the Product Owner and stakeholders. Dependencies were also considered — for example, assessment-related stories (EP-02) must precede AI analysis stories (EP-03), which in turn must precede recommendation stories (EP-04–EP-06). The table below lists the full backlog ranked by business value, MoSCoW priority and story points.

Rank	ID	User Story (Summary)	MoSCoW	Value	Points
1	US-04	As a learner, I want to take a technical skill assessment relevant to my domain, so that my ...	Must	10	8
2	US-10	As a learner, I want to receive a ranked list of recommended career paths, so that I can exp...	Must	10	13
3	US-01	As a new learner, I want to register using email or social login, so that I can quickly crea...	Must	9	3
4	US-05	As a learner, I want to take a soft-skill assessment (communication, teamwork, problem solvi...	Must	9	8
5	US-08	As a learner, I want the system to analyse my aptitude using AI, so that I understand my nat...	Must	9	13
6	US-13	As a learner, I want my skills compared against current industry/job requirements, so that I...	Must	9	8
7	US-02	As a learner, I want to build a profile with my education, interests and career goals, so th...	Must	8	5
8	US-14	As an industry partner, I want to publish up-to-date skill requirements for open roles, so t...	Must	8	8
9	US-16	As a learner, I want to receive suggested courses and certifications for my skill gaps, so t...	Must	8	8
10	US-19	As a learner, I want to generate a personalised career report as a PDF, so that I can share ...	Must	8	5
11	US-06	As a learner, I want the assessment to adapt its difficulty based on my answers, so that the...	Should	7	13
12	US-11	As a learner, I want to see why a particular career was recommended to me, so that I can tru...	Should	7	8
13	US-15	As a learner, I want to see the skill gap between my current profile and a target job role, ...	Should	7	5
14	US-20	As a career counsellor, I want to view a learner's full report with assessment and recommend...	Should	7	5
15	US-21	As a learner, I want to track my learning progress on a dashboard, so that I stay motivated ...	Should	7	8
16	US-23	As an administrator, I want to add, edit or remove assessment questions, so that content sta...	Must	7	8
17	US-09	As a learner, I want my preferred learning style (visual, auditory, kinesthetic) identified,...	Should	6	8

Rank	ID	User Story (Summary)	MoSCoW	Value	Points
18	US-17	As a learner, I want to filter suggested courses by cost, duration and mode (online/offline)...	Should	6	5
19	US-24	As an administrator, I want to monitor platform usage and assessment analytics, so that I ca...	Should	6	5
20	US-25	As an administrator, I want to manage industry-partner data feeds, so that job-market inform...	Should	6	5
21	US-03	As a learner, I want to reset my forgotten password securely, so that I can regain access to...	Should	5	2
22	US-12	As a learner, I want to compare two or more recommended career paths side by side, so that I...	Could	5	5
23	US-18	As a training provider, I want my certification programs listed against relevant skill gaps,...	Could	5	5
24	US-22	As a learner, I want to retake assessments periodically, so that my recommendations stay upd...	Could	5	5
25	US-07	As a learner, I want to save and resume an in-progress assessment, so that I do not lose my ...	Could	4	5
26	US-26	As a learner, I want to receive email/app notifications about incomplete assessments or new ...	Could	4	3
27	US-27	As a learner, I want to receive alerts when a new job or certification matches my profile, s...	Could	4	5

4.1 Prioritization Rationale

- Onboarding and core assessment stories (US-01, US-02, US-04, US-05) rank highest because no downstream AI analysis or recommendation can occur without learner profile and assessment data — they are hard dependencies for every other epic.
- AI aptitude analysis (US-08) and the career recommendation engine (US-10) carry the highest business value as they represent the platform's core differentiating value proposition.
- Industry skill-matching (US-13, US-14) and training recommendations (US-16) were prioritised next since they directly drive the 'employability' outcome central to the project vision.
- Reporting (US-19) was included in Sprint 1 as a Must because a visible, shareable output is required to demonstrate end-to-end value in the very first release.
- Engagement features (notifications, retake reminders) and comparison/filtering conveniences were ranked lower (Could) as they enhance but are not essential to the core value loop.

5. Sprint Backlog Plan (Sprint 1)

Sprint Duration: 2 weeks (10 working days)

Team Capacity: 1 Product Owner, 1 Scrum Master, 2 Backend Developers, 1 Frontend Developer, 1 AI/ML Engineer, 1 QA Engineer, 1 UI/UX Designer (part-time), 1 Subject Matter Expert (part-time)

Sprint Capacity (Story Points): 62 points (based on historical/estimated team velocity)

The nine highest-priority user stories from the top of the prioritised backlog were pulled into Sprint 1, totalling 61 story points — within the team's estimated capacity of 62 points. These stories deliver one complete, end-to-end “thin slice” of the product: a learner can register, build a profile, take both assessments, receive an AI aptitude and career recommendation, see an industry skill match, get a course suggestion, and download a report.

5.1 Selected Sprint 1 Stories

ID	User Story (Summary)	Points
US-01	As a new learner, I want to register using email or social login, so that I can quickly create an account a...	3
US-02	As a learner, I want to build a profile with my education, interests and career goals, so that recommendati...	5
US-04	As a learner, I want to take a technical skill assessment relevant to my domain, so that my current compete...	8
US-05	As a learner, I want to take a soft-skill assessment (communication, teamwork, problem solving), so that no...	8
US-08	As a learner, I want the system to analyse my aptitude using AI, so that I understand my natural strengths ...	13
US-10	As a learner, I want to receive a ranked list of recommended career paths, so that I can explore options th...	13
US-13	As a learner, I want my skills compared against current industry/job requirements, so that I know how job-r...	8
US-16	As a learner, I want to receive suggested courses and certifications for my skill gaps, so that I can plan ...	8
US-19	As a learner, I want to generate a personalised career report as a PDF, so that I can share it with counsel...	5

5.2 Task Breakdown

Each selected user story was broken down into actionable development tasks, assigned to the appropriate role, with hour-level estimates to support daily stand-up tracking.

US-01: As a new learner, I want to register using email or social login, so that I can quickly create an account and start my assessment.

Task	Assigned To	Est. Hours
Design registration UI (email + social login)	Frontend Developer	6
Implement registration API & validation	Backend Developer	8
Integrate email verification service	Backend Developer	5
Write unit & integration tests	QA Engineer	4

Total estimated effort: 23 hours

US-02: As a learner, I want to build a profile with my education, interests and career goals, so that recommendations are personalised to me.

Task	Assigned To	Est. Hours
Design profile-setup form (education, interests, goals)	Frontend Developer	6
Build profile data model & API	Backend Developer	6
Add profile completeness validation logic	Backend Developer	3
Test profile CRUD operations	QA Engineer	3

Total estimated effort: 18 hours

US-04: As a learner, I want to take a technical skill assessment relevant to my domain, so that my current competency level is measured accurately.

Task	Assigned To	Est. Hours
Curate/domain technical question bank	Content/Subject Matter Expert	10
Build assessment engine (timer, scoring)	Backend Developer	10
Design assessment UI with progress indicator	Frontend Developer	6
Test scoring accuracy across domains	QA Engineer	5

Total estimated effort: 31 hours

US-05: As a learner, I want to take a soft-skill assessment (communication, teamwork, problem solving), so that non-technical strengths are also captured.

Task	Assigned To	Est. Hours
Curate soft-skill scenario question set	Content/Subject Matter Expert	8
Build soft-skill scoring logic	Backend Developer	6
Design soft-skill assessment UI	Frontend Developer	5
Test resume-assessment functionality	QA Engineer	3

Total estimated effort: 22 hours

US-08: As a learner, I want the system to analyse my aptitude using AI, so that I understand my natural strengths and reasoning ability.

Task	Assigned To	Est. Hours
Select/train AI aptitude analysis model	AI/ML Engineer	14
Build API to serve aptitude results	Backend Developer	6
Design aptitude visualisation (charts)	Frontend Developer	6
Validate model output against test data set	QA/Data Analyst	6

Total estimated effort: 32 hours

US-10: As a learner, I want to receive a ranked list of recommended career paths, so that I can explore options that fit my skills and interests.

Task	Assigned To	Est. Hours
Design career-matching algorithm (rules + ML)	AI/ML Engineer	14
Build recommendation API & ranking logic	Backend Developer	8
Design recommendation results UI	Frontend Developer	6
Test recommendation relevance with sample profiles	QA Engineer	6

Total estimated effort: 34 hours

US-13: As a learner, I want my skills compared against current industry/job requirements, so that I know how job-ready I am.

Task	Assigned To	Est. Hours
Integrate industry skill-demand data feed	Backend Developer	8
Build skill-gap comparison logic	AI/ML Engineer	8
Design job-readiness score UI	Frontend Developer	5
Test matching accuracy with sample job roles	QA Engineer	4

Total estimated effort: 25 hours

US-16: As a learner, I want to receive suggested courses and certifications for my skill gaps, so that I can plan my learning path.

Task	Assigned To	Est. Hours
Build course/certification recommendation mapping	Backend Developer	8
Integrate training-provider course catalogue	Backend Developer	6
Design 'Recommended Courses' UI with filters	Frontend Developer	5
Test course mapping against known gaps	QA Engineer	3

Total estimated effort: 22 hours

US-19: As a learner, I want to generate a personalised career report as a PDF, so that I can share it with counsellors, parents or employers.

Task	Assigned To	Est. Hours
Design PDF report template	UI/UX Designer	5
Build report-generation service	Backend Developer	8
Add download/share functionality	Frontend Developer	4
Test report accuracy & performance	QA Engineer	4

Total estimated effort: 21 hours

6. Story Point Estimation

Story points were estimated using Planning Poker with the Fibonacci sequence (1, 2, 3, 5, 8, 13, 21...), a technique chosen because it forces relative — rather than absolute — sizing and surfaces disagreement early through discussion. Each team member privately selected a card representing their estimate for a story; outliers discussed their reasoning; the team re-voted until consensus was reached. Estimates reflect complexity, uncertainty/risk and effort combined, not calendar time alone.

6.1 Estimation Scale and Justification

Points	Meaning / Complexity Level	Applied To (examples)
1 - 2	Trivial, well-understood UI text/label changes or minor config updates.	N/A this sprint
3	Small, low-risk feature with clear requirements and no external dependency.	US-01, US-26
5	Moderate feature involving a new data model, form, or integration with one dependency.	US-02, US-03, US-15, US-17, US-19, US-24, US-25
8	Complex feature requiring backend + frontend + integration work, or moderate AI/analytics logic.	US-04, US-05, US-09, US-11, US-13, US-14, US-16, US-21, US-23
13	High complexity involving AI/ML model design, training or an adaptive algorithm with uncertain edge cases.	US-06, US-08, US-10
21+	Very large, spans multiple sub-systems; must be split before entering a sprint.	None identified (backlog kept below 21)

6.2 Full Backlog Estimation Summary

ID	Epic	Points	Justification
US-01	EP-01	3	Small, isolated change
US-02	EP-01	5	Moderate, one dependency
US-03	EP-01	2	Trivial
US-04	EP-02	8	Complex, multi-layer feature
US-05	EP-02	8	Complex, multi-layer feature
US-06	EP-02	13	High complexity / AI logic
US-07	EP-02	5	Moderate, one dependency
US-08	EP-03	13	High complexity / AI logic
US-09	EP-03	8	Complex, multi-layer feature
US-10	EP-04	13	High complexity / AI logic

ID	Epic	Points	Justification
US-11	EP-04	8	Complex, multi-layer feature
US-12	EP-04	5	Moderate, one dependency
US-13	EP-05	8	Complex, multi-layer feature
US-14	EP-05	8	Complex, multi-layer feature
US-15	EP-05	5	Moderate, one dependency
US-16	EP-06	8	Complex, multi-layer feature
US-17	EP-06	5	Moderate, one dependency
US-18	EP-06	5	Moderate, one dependency
US-19	EP-07	5	Moderate, one dependency
US-20	EP-07	5	Moderate, one dependency
US-21	EP-08	8	Complex, multi-layer feature
US-22	EP-08	5	Moderate, one dependency
US-23	EP-09	8	Complex, multi-layer feature
US-24	EP-09	5	Moderate, one dependency
US-25	EP-09	5	Moderate, one dependency
US-26	EP-10	3	Small, isolated change
US-27	EP-10	5	Moderate, one dependency

Total Product Backlog Story Points: 179

Sprint 1 Committed Story Points: 61 (of 62 available capacity)

Estimated Remaining Sprints to Complete Backlog: 3 additional sprints, assuming a steady-state velocity of ~55 points/sprint once the team stabilises

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

“By the end of Sprint 1, a learner shall be able to register, complete a technical and soft-skill assessment, receive an AI-generated aptitude summary and at least one ranked career recommendation with an industry skill-match score, view a suggested course for their top skill gap, and download a personalised career report — demonstrating one complete, working slice of the platform end-to-end.”

7.2 Expected Sprint Deliverables

- A working registration, login and profile-setup flow (US-01, US-02).
- A functioning technical skill assessment module with domain-specific scoring (US-04).

- A functioning soft-skill assessment module with category-wise scoring (US-05).
- An AI aptitude analysis engine producing a strengths/improvement summary (US-08).
- A career recommendation engine returning a ranked list of career matches (US-10).
- An industry skill-matching feature showing a job-readiness percentage (US-13).
- A course/certification recommendation list mapped to the learner's top skill gap (US-16).
- A downloadable PDF career report consolidating all Sprint 1 outputs (US-19).
- Automated unit/integration test coverage for all Sprint 1 stories, and a sprint demo to stakeholders.

7.3 Definition of Done (DoD)

- Code is peer-reviewed, merged to the main branch and passes CI build checks.
- Unit and integration tests are written and passing with no critical/blocker defects.
- Acceptance criteria for the story are verified and signed off by the Product Owner.
- UI is responsive and matches agreed design mock-ups.
- Feature is deployed to the staging environment and demonstrated at the Sprint Review.
- Relevant documentation (API docs, user-facing help text) is updated.

7.4 Sprint Ceremonies Plan

Ceremony	Frequency / Timing	Purpose
Sprint Planning	Day 1 (4 hours)	Select backlog items, define sprint goal, break stories into tasks
Daily Stand-up	Every working day (15 min)	Sync on progress, blockers and plan for the day
Backlog Refinement	Mid-sprint, Day 5 (1 hour)	Groom and estimate upcoming backlog items for next sprint
Sprint Review / Demo	Last day (1.5 hours)	Demonstrate completed increment to stakeholders, gather feedback
Sprint Retrospective	Last day (1 hour)	Reflect on process, identify improvements for next sprint

7.5 Risks and Mitigations

Risk	Impact	Mitigation
AI aptitude/recommendation model accuracy is low with limited training data	High	Start with a rule-based + lightweight ML hybrid model; refine with real usage data post-launch
Industry skill-demand data feed is delayed or incomplete	Medium	Use a curated seed dataset for Sprint 1; integrate live feeds incrementally
Assessment content (question bank) is not ready in time	Medium	Engage Subject Matter Expert from Day 1; prioritise one domain for MVP
Scope creep from stakeholders during the sprint	Medium	Product Owner protects the sprint goal; new requests go to backlog for next sprint

7.6 Conclusion

This sprint plan establishes a clear, prioritised path from the problem statement to a working, demonstrable increment of the AI-Powered Smart Vocational Skill Assessment and Career Recommendation Platform. By focusing Sprint 1 on the complete assessment-to-recommendation-to-report journey, the team delivers early, tangible value while laying the technical foundation — profile data, assessment scoring, AI models and industry data integration — needed for all subsequent sprints covering progress monitoring, admin tooling and engagement features.

The Agile approach adopted throughout this workshop ensures that the platform is built incrementally, with each sprint producing a usable, testable increment rather than a large, high-risk release at the end of the project. Breaking the problem statement into ten well-scoped epics and twenty-seven independently valuable user stories allows the Product Owner to reorder the backlog at any time in response to changing stakeholder needs, new industry data, or feedback gathered during Sprint Reviews, without disrupting work already in progress.

Writing Given–When–Then acceptance criteria for every Sprint 1 story before development begins reduces ambiguity between the Product Owner and the development team, minimises rework caused by misunderstood requirements, and gives QA a ready-made basis for test-case design. Combined with a shared Definition of Done, this creates a consistent, measurable standard of quality across every story delivered in the sprint.

Using Planning Poker with the Fibonacci sequence for story-point estimation also builds a shared, evolving understanding of complexity within the team. As Sprint 1 is delivered and actual effort is compared against estimates, the team's velocity will stabilise, allowing more accurate sprint forecasting and release planning for the remaining backlog in future sprints.

Overall, this sprint planning document demonstrates how core Agile practices — backlog prioritisation, sprint planning, story-point estimation and a clearly stated sprint goal — can be applied end-to-end to a real-world problem statement, turning a broad vision of AI-driven career guidance into a concrete, achievable, and continuously improvable delivery plan.